Data Structures and Algorithms · 21CSC201J

Unit 1 · Session 7 · SLO 1 — Solved Worksheet

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Unit â€“ 1 : Session â€“ 7 : SLO â€“ 1 | Data Structures â€” Definition and Types

I. Fill in the Blanks

1. data structure â€” a way of organizing and storing data for efficient access and modification.
2. linear and non-linear â€” the two broad types.
3. linear â€” elements arranged sequentially.
4. Queue â€” the FIFO linear data structure.
5. Tree â€” the non-linear structure for hierarchical relationships.
6. graph â€” elements point to multiple elements, forming a network.
7. push and pop â€” the two stack operations.
8. array â€” a collection of elements identified by index.
9. linked list â€” nodes containing data plus a next pointer.
10. storage (organization) and retrieval (processing) â€” efficient management of data.

II. Match the Following

1. Array â€” C. Sequential memory allocation.
2. Stack â€” A. Last In First Out.
3. Queue â€” B. First In First Out.
4. Linked List â€” D. Each node points to the next node.
5. Tree â€” J. Hierarchical structure.
6. Graph â€” H. Represents complex relationships.
7. Linear Data Structure â€” F. Stack (example).
8. Non-Linear Data Structure â€” I. Graph (example).
9. Stack Operations â€” G. Push and Pop.
10. Queue Operations â€” E. Enqueue and Dequeue.